

Szymon Jakubicz

Machine Learning Engineer
Data Scientist

CONTACT

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Warsaw, Poland

SKILLS

Machine Learning
PyTorch, CNNs, GNNs, computer vision, OpenCV, NLP, statistical learning, reinforcement learning, multi-object tracking (FairMOT)

LLMs & GenAI
information extraction, RAG, AI agents, OpenAI API, Llama, Qwen

Knowledge Graphs
graph data modeling, entity and relationship extraction, Cypher

Edge AI
NVIDIA Jetson AGX Xavier, real-time inference, CUDA

Software Engineering
Python, JavaScript, TypeScript, React, Node.js, REST APIs, SQL, Postgres, Git, testing, Docker

Mathematics
linear algebra, probability and statistics, numerical methods

LANGUAGES

Polish - native

English - fluent, professional working proficiency

SUMMARY

Machine Learning Engineer and Data Scientist with 4+ years of production experience building end-to-end AI systems: LLM-powered pipelines that turn unstructured documents into knowledge graphs, computer vision models deployed on NVIDIA Jetson edge devices, and full-stack products around them. Works directly with Polish and international clients to turn open-ended questions into production-ready systems. First author of an arXiv paper on reinforcement learning for vehicle routing.

EXPERIENCE

Data Scientist - Million Monkeys Software

July 2022 - Present · Warsaw, Poland

- Designed and built an end-to-end pipeline converting unstructured documents (PDFs, scanned images, CSVs) into knowledge graphs, using LLMs (Llama, Qwen) for entity extraction and relationship inference, deployed in production for client projects.
- Benchmarked multi-object tracking models for speed and accuracy on NVIDIA Jetson AGX Xavier; selected FairMOT and, as it exceeded the device's real-time capacity, designed a hybrid architecture combining FairMOT with physics-based motion prediction, achieving real-time tracking while maintaining detection accuracy.
- Developed a text editor with an embedded AI writing agent for a publishing house, used daily by its journalists; worked across the frontend (JavaScript, React), backend and the LLM agent as the client's main point of contact from requirements to delivery.
- Built a full-stack web platform and REST APIs (JavaScript/React, FastAPI) for internal teams and clients to explore pipeline outputs and knowledge graphs.
- Led requirements calls with Polish and international clients, turning business needs into technical specifications and client-facing documentation.
- Reviewed code across the engineering team and used AI-assisted development (code assistants, automated review) in daily work.
- Joined as an undergraduate student and grew into end-to-end ownership of projects, from research question to production.

PUBLICATIONS

POMO+: Leveraging Starting Nodes in POMO for Solving Capacitated Vehicle Routing Problem

S. Jakubicz et al. (first author) · arXiv, 2025 · arxiv.org/abs/2508.08493

- Extended POMO, a state-of-the-art reinforcement learning solver for vehicle routing, by introducing an attention-based agent selecting starting nodes; achieved faster convergence and lower optimality gaps on CVRPLIB benchmarks (PyTorch, RL4CO).

EDUCATION

Postgraduate Studies, Cybersecurity and Digital Forensics (in progress)

Lazarski University · 2026 - present

MSc, Machine Learning and Statistical Methods

University of Warsaw · 2023 - 2025

- Thesis: **Graph-based disinformation detection using graph neural networks (GNNs)**

BSc, Computer Science

University of Warsaw · 2020 - 2023

- Thesis: **System do budowy modeli agentowych - Interfejs użytkownika**

BSc, Mathematics

University of Warsaw · 2020 - 2024

- Thesis: **Wykorzystanie grafowych sieci neuronowych w celu przewidywania rezygnacji uczestników kursów internetowych**

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